

考虑加速衰减特性的燃料电池压缩机性能预测研究

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研究对象：燃料电池压缩机

研究目标：通过设计加速耐久性试验，获取带空气轴承的燃料电池离心式空气压缩机在加速衰减条件下的性能数据，明确其性能衰减规律。

数据来源：来自于课题组内部燃料电池空压机实验数据。

研究方法：采用“外层优化+内层训练”的双层结构-外层以集成学习模型在测试集上的均方误差（MSE）为目标，通过 HyVABC 算法优化模型超参数；内层根据指定超参数训练集成学习模型（含随机森林 RF、自适应提升 AdaBoost、极限梯度提升 XGBoost 三种），输入为压缩机运行参数（磨合时间、转速、功率），输出为性能参数（压比、流量），并添加“测试集 MSE 大于训练集 MSE”的约束以避免过拟合。分析三种组合模型（RF-HyVABC、AdaBoost-HyVABC、XGBoost-HyVABC）的探索率与利用率、迭代过程中测试集 MSE 变化、每代训练时间，对比预测值与试验值的相对误差，筛选最优模型。

主要结果：（1）XGBoost-HyVABC 模型综合表现最优-每代训练时间最短（46 s），测试集 MSE 最小（16.4），探索率与利用率动态平衡能力最强，能更高效地利用已知优质解加速收敛。（2）转速 100000 rpm、流量 133 g/s 时，压比衰减程度最大，较磨合前衰减 25.9%。

Study on Performance Prediction of Fuel Cell Compressor Considering Accelerated Degradation Characteristics

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Abstract

As a core component of the fuel cell air supply system, the air compressor plays a crucial supporting role in the efficient and stable operation of the fuel cell system. This study conducts accelerated durability tests on air compressors equipped with air bearings based on preset operating parameters. At the same time, a Hybrid Variational Artificial Bee Colony (HyVABC) algorithm is proposed to enhance the algorithm's global optimization capability and local feature capturing ability in a collaborative manner, achieving a dynamic balance. On this basis, a combined prediction model is further constructed to accurately predict the performance of the compressor considering the accelerated degradation characteristics. The experimental and analytical results indicate that when the compressor flow is 133 g/s and the speed is 100,000 rpm, the degree of pressure ratio degradation is most significant, with a degradation rate of 25.9% compared to the initial state. Among 20 test problems, the HyVABC algorithm found the global optimal solution in 16 cases, highlighting its excellent optimization performance, fast convergence rate, and high robustness. The XGBoost-HyVABC model was selected to establish the mapping relationship between compressor operating parameters and performance parameters. The relative error of the predicted values and experimental measurements is controlled within an absolute value of 3%. These research results provide an efficient heuristic optimization algorithm and high-precision combined prediction model for performance prediction under accelerated degradation characteristics of fuel cell compressors, offering theoretical and technical support for future compressor performance regulation and life assessment.

Keywords: Fuel cell; Centrifugal air compressor; HyVABC algorithm; Combinatorial prediction model; Accelerated decay characteristics.

1. Introduction

The widespread use of fossil fuels has led to the emission of large amounts of harmful gases [1], causing serious environmental pollution and having a profound negative impact on ecosystems [2]. Therefore, the development of efficient and environmentally friendly new energy vehicles is of unprecedented significance [3]. Among them, hydrogen fuel cell vehicles (HFCVs), with their high efficiency and zero emissions, have become a sustainable solution to reduce emissions related to fossil fuels [4].

Centrifugal air compressors are core components of fuel cell systems. By pressurizing the intake air, they enhance power density [5]. Their performance directly affects the system's stability and efficiency, necessitating detailed analysis and prediction [6]. Currently, methods such as one-dimensional (1-D), three-dimensional (3-D), and machine learning (ML) have been employed in the academic field to establish performance prediction models for compressors [7].

Existing one-dimensional models are constructed based on theoretical frameworks and engineering experience, calibrated with experimental data [8]. For example, Chen et al. [9] used a one-dimensional loss model to identify the key performance-influencing parameters, namely the impeller outlet width and radius. Xia et al. [10] coupled one-dimensional and CFD models, improving the compressor's adiabatic efficiency to over 90%. These models simulate quickly but rely heavily on experience, resulting in some loss of accuracy. Three-dimensional models are generally based on experimental data and are used for performance prediction, parameter optimization, and structural design [11]. Ekradi et al. [12] optimized the geometry of impellers with splitter blades using a three-dimensional model, enhancing overall performance. Zhang et al. [13] incorporated Laval nozzle methods to accurately predict surge characteristics under supersonic conditions. Three-dimensional models have strong predictive capabilities but require complex calibration and have high computational costs, limiting their application. Machine learning models are built based on large amounts of experimental and simulation data [14]. Ding et al. [15] combined genetic algorithms, backpropagation neural networks, and support vector machines to construct a hybrid model. Hosseinimaab et al. [16] optimized the impeller through neural networks, achieving a 29.62% increase in pressure ratio. These models have high accuracy and fast computation but rely

on massive datasets and suffer from frequent manual parameter adjustments, time-consuming training, and overfitting issues.

Accurate and efficient prediction models are the foundation for evaluating and optimizing compressor performance [17]. However, current 1-D, 3-D, and ML models generally face issues such as low accuracy, complex computation, and large data requirements (see Table 1), and they do not adequately account for the effects of compressor degradation, particularly accelerated degradation. Due to the high cost of accelerated degradation testing and the difficulty in obtaining data, constructing related prediction models is challenging. Therefore, this study develops a combined prediction model based on accelerated durability test data, integrating ensemble learning with the HyVABC algorithm, to achieve accurate prediction of compressor accelerated degradation performance. The main work is as follows:

Table 1. Comparisons of advantages and disadvantages between 1-D, 3-D and ML models.

Type	Advantage	Disadvantage
1-D model	Simple; easy to use; fast computation	Limited accuracy; limited predictive power for complex systems or high-accuracy needs
3-D model	High-accuracy prediction; applicable to complex systems and design optimization	High computational cost; high complexity; strong dependence on experimental data
ML model	High-accuracy prediction; fast computation; high degree of automation	Strong data dependence; limited generalization ability; lack of interpretability; risk of overfitting

2 Method

2.1 Accelerated endurance test

The test subject is a centrifugal air compressor powered by an electric motor with a rated speed of 120,000 rpm. It uses two-stage compression and is water-cooled, with a coolant mixture of ethylene glycol and water. The compressor's structure and interface are shown in Fig. 1, with key specs in Table 2. It mainly consists of the rotor, stator, and control elements. The rotor includes the impeller, shaft, and bearing, which generate compressed air power through rotation. The stator contains the inlet chamber, diffuser, and outlet chamber, forming the compressed air pathways. The

control element includes the drive and regulation systems that manage rotor speed and air pressure.

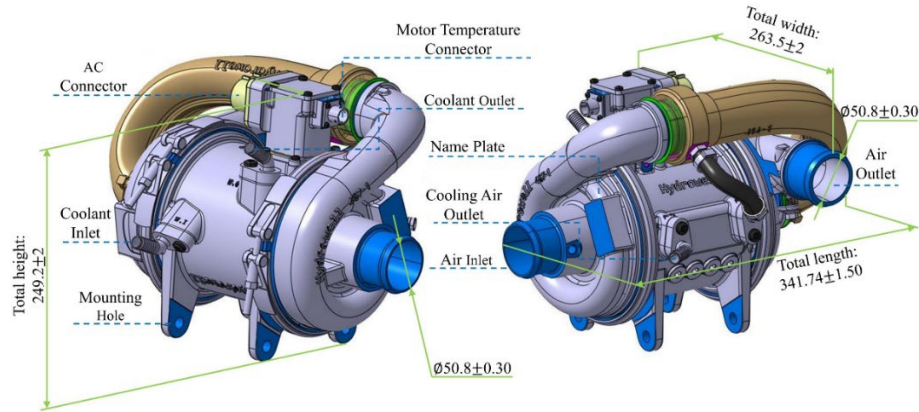


Fig. 1. Geometric structure and interface arrangement of the compressor.

Table 2. Key technological specifications of the compressor.

Item	Content
Speed range (rpm)	30000 ~ 120000
Maximum mass flow rate (g/s)	165
Operating ambient temperature (°C)	-40 ~ 65
Operating ambient humidity (%RH)	5 ~ 99
Maximum outlet temperature (°C)	200
Type (-)	centrifugal
Rated current (A)	50
Full power operating voltage range (V)	450 ~ 750
Rated power of motor (kW)	22

2.2 Ensemble learning algorithms

Ensemble learning algorithms build stronger and more robust models by combining multiple weak learners, with the core idea being to improve the overall performance by integrating the predictions of multiple models [18]. The two main categories of integrated learning are Bagging and Boosting algorithms, and Random Forest (RF) is a strong Bagging algorithm which improves the model performance by incorporating prediction results of several decision trees [19], as depicted in Fig. 2. The core idea is to achieve stochasticity and diversity for the model by self-sampling the training dataset and randomly selecting the features of each decision tree.

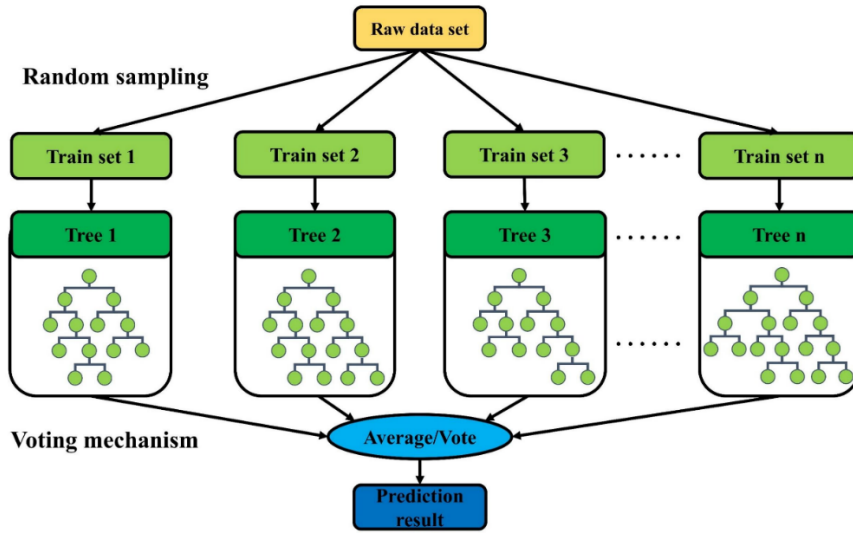


Fig. 2. Flowchart of the RF algorithm.

Adaptive Boosting (AdaBoost) is a classical Boosting algorithm whose goal is to build stronger learners by iteratively training multiple weak learners [20], and the flowchart is shown in Fig. 3. The basic idea of AdaBoost is to assign different weights to distinct samples in the training set, so that those samples misclassified by previous classifications can draw more attention in the subsequent training. The training process of AdaBoost starts with assigning equal weights to all samples, then adjusting the weights of the samples by training weak learners. Within each iteration, AdaBoost tweaks the weights of the samples so that more attention will be paid to the samples previously misclassified in the next round to gradually improve the overall performance. The weights of weak learners are adjusted upon their performance on the training set with better or worse performing learners receiving larger or smaller weights, respectively.

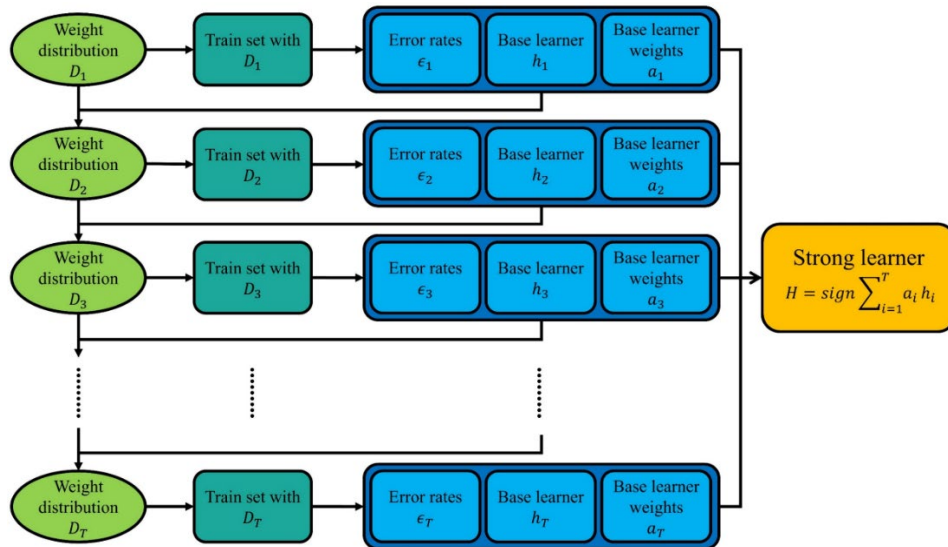


Fig.3. Flowchart of the AdaBoost algorithm.

The eXtreme Gradient Boosting (XGBoost) is a Boosting algorithm relying on the gradient boosting framework, which is introduced with the regularization term and second-order derivative information [21], as illustrated in Fig. 4. In the process of structuring each tree, a special loss function is utilized in XGBoost that incorporates an approximate gradient of the predicted values to fit the residuals more accurately. The centerpiece of XGBoost includes the flexible control of the tree model, adjusting the model complexity through regularization and enabling dynamic feature selection at the leaf nodes. The gradient information is imported into XGBoost in each iteration to realize the adjustment of adaptive learning rate, which makes the model more adaptable to different data distributions, thus improving the stability and convergence speed of training.

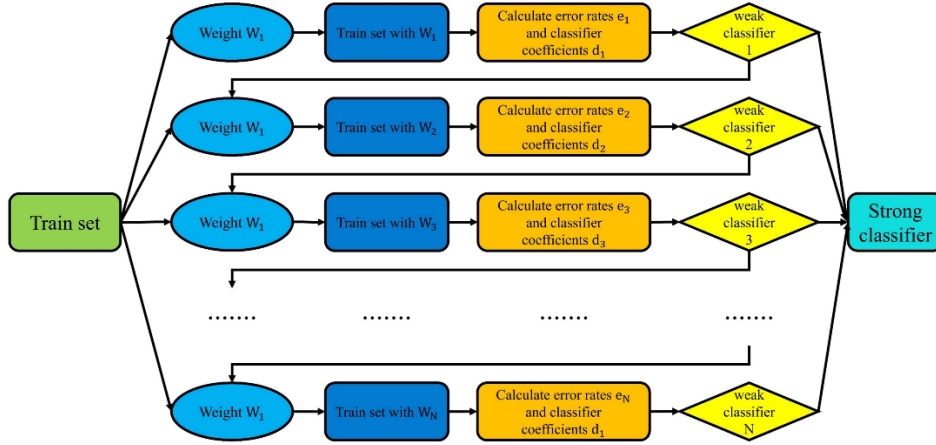


Fig. 4. Flowchart of the XGBoost algorithm.

The Artificial Bee Colony (ABC) algorithm emulates the search strategy employed by honeybees when seeking food, with the ultimate objective of discovering the best solution for a given problem [22]. The ABC algorithm is motivated by collaborative feeding behaviors of honeybee colonies, allowing for efficient location of food sources within expansive search spaces through collaboration and information transfer. The essential procedure of the ABC algorithm encompasses initialization, employed bees, onlookers, scouts and updating phases, where the search equation, selection probability and random generation strategy are illustrated in Eqs. (1–3).

$$v_{ij} = x_{ij} + \phi_{ij}(x_{ij} - x_{kj}) \quad (1)$$

$$P_i = \frac{f(X_i)}{\sum_{n=1}^N f(X_n)} \quad (2)$$

$$x_{ij} = x_j^{min} + rand(0,1)(x_j^{max} - x_j^{min}) \quad (3)$$

where k and j are randomly selected subscripts, $k \neq i$ and $k \in \{1, 2, \dots, N\}$, $j \in \{1, 2, \dots, D\}$; N is individual number of the population; D is the dimension of each individual vector; P_i is the

probability of the onlookers choosing food sources; f is the objective function; φ_{ij} is a random number between -1 and 1 ; x_j^{min} and x_j^{max} are the lower and upper bounds of the search space, respectively; $rand(0,1)$ represents a random number between 0 and 1 .

Although the ABC algorithm excels in some aspects, there exist some challenges when dealing with high-dimensional, nonlinear and large-scale problems, like prone to local optimums, unsteady convergence rate and insufficient global search ability. To strike a great balance between global optimization ability and local feature capturing capability, the HyVABC algorithm is proposed, which is inspired by the differential evolutionary (DE) algorithm [23] and the particle swarm optimization (PSO) algorithm [24]. The HyVABC algorithm improves search equations of the employed bees and onlookers, as illustrated in Eqs (4–5). With the improvements, the employed bees focus more on local features when updating, while the onlookers concentrate more on global features, thus achieving an enhancement and balance between global optimization ability and local feature capturing capability.

$$v_{ij} = x_{ij} + \varphi_{ij}(x_{best,j} - x_{ij}) + \varphi_{ij}(x_{best,j} - x_{kj}) \quad (4)$$

$$v_{ij} = x_{best,j} + \varphi_{ij}(x_{best,j} - x_{ij}) + \varphi_{ij}(x_{best,j} - x_{kj}) \quad (5)$$

where φ_{ij} is a random value between -1 and 1 ; x_{best} is the best individual of current population.

2.3 The combinatorial prediction model

The hyperparameter tuning of the ensemble learning model is a complex nonlinear optimization problem, and manual tuning alone is hard to achieve the best model performance. Therefore, the construction and tuning of the ensemble learning model is summarized as a combinatorial optimization problem by combining the HyVABC algorithm, as illustrated in Eq. (6):

$$\text{The external layer: } \min \frac{1}{L_{test}} \sum_{i=1}^{L_{test}} (y_i^* - y_i)^2$$

$$\text{The internal layer: } y^* = E(T_b, S_A, P_A; N_C, \varepsilon/M_L, M_D) = [\sigma, f_r]$$

$$\begin{aligned} \frac{1}{L_{test}} \sum_{i=1}^{L_{test}} (y_i^* - y_i)^2 &> \frac{1}{L_{train}} \sum_{i=1}^{L_{train}} (y_i^* - y_i)^2 \\ N_C &\in [1, 200] \\ \varepsilon &\in [0.001, 1.5] \\ M_L &\in [2, 50] \\ M_D &\in [1, 25] \\ S_A, N_C, M_L, M_D &\in N^* \\ T_b, P_A, \varepsilon &\in R \end{aligned} \quad (6)$$

where L_{test} and L_{train} are data volumes of the training and test sets, respectively; E denotes the ensemble learning model; T_b , S_A and P_A are the break-in time, speed and power of the air compressor, respectively. σ and f_r are pressure ratio and flow rate, respectively; N_C , ε , M_L and M_D are CART number, learning rate, maximum leaf node and depth of CARTs, respectively.

The mapping relationship between compressor operating parameters and performance parameters is described by constructing combinatorial prediction models, as demonstrated in Fig.5. In external layer, the mean square error (MSE) [25] between predicted values of ensemble learning models and experimental values on test set is taken as the target, and the HyVABC algorithm is employed for optimization. The smaller the MSE, the more superior prediction performance of combinatorial prediction models. In internal layer, the ensemble learning model is trained according to the specified model hyperparameters to ensure that the model can accurately fit the training set. The inputs of the ensemble learning models are three operating parameters of the compressor, and outputs are flow rate and pressure ratio. Meanwhile, the constraint that MSE of combinatorial prediction models on test set always remains larger than that on training set is added. This design idea is based on the property that the combinatorial prediction model has higher predictive ability for known data (the training set) and lower predictive ability for unknown data (the test set). Therefore, imposing this constraint can effectively prevent the ensemble learning model from underfitting and avoid the optimization process from being too biased towards minimizing MSE on test set.

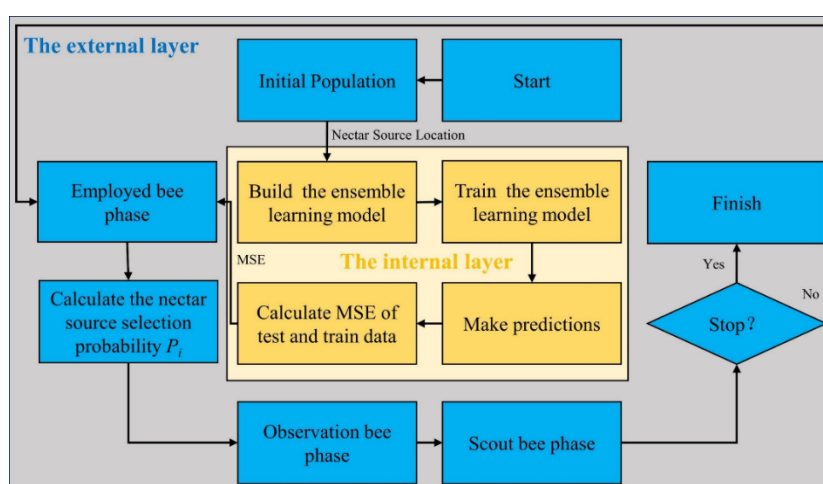


Fig. 5. Specific structure of the combinatorial prediction model.

3 Results and discussions

3.1 Analysis of accelerated decay characteristics

The pressure ratio-flow curves of the air compressor at various speeds before and after break-in are illustrated in Fig. 6, where cases 1–7 indicate different speed conditions (50000–110000 rpm, with a 10000 rpm interval) before break-in and cases 8-14 represent those after break-in. The pressure ratio of the air compressor after break-in at different speeds is smaller than that before break-in when the flow rate is the same, which demonstrates a significant decay of the performance for the compressor after break-in. Meanwhile, the decay degree of the air compressor first increases and then decreases as the speed increases, which is attributed to the fact that the air compressor is more operated in the medium speed condition. In addition, the accelerated decay characteristic of the air compressor with a speed of 100000 rpm is focused on. After 60 h of break-in, the decay degree of pressure ratio is the greatest when the flow rate is 133 g/s, with the decay rate of 25.9 % compared with that before break-in.

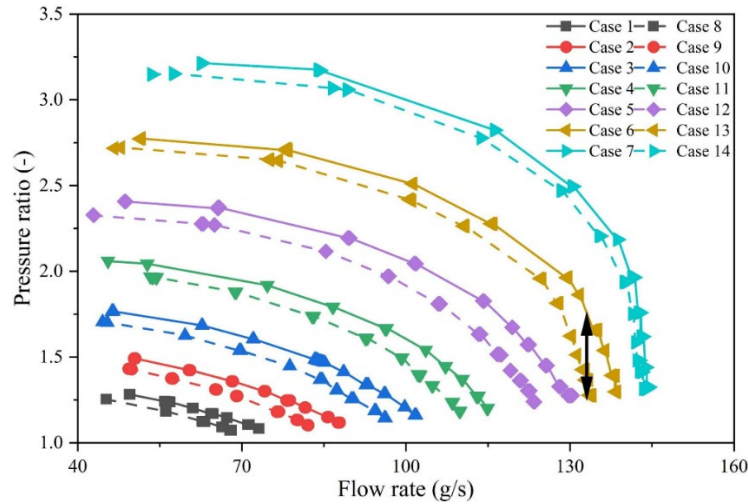


Fig. 6. Pressure ratio-flow curves for various accumulated break-in times.

3.2 Performance of the HyVABC algorithm

To verify the performance of the HyVABC algorithm, search equations of the ABC algorithm for employed bees and onlookers are improved with Eq. (4) and Eq. (5) individually to form the ABC1 and ABC2 algorithms. Twenty test problems [26] are selected to test the performance of the ABC, ABC1, ABC2 and HyVABC algorithms. To ensure the validity and generality of results, 30 optimizations are conducted independently for all test problems, and the mean of all optimal results are counted. The same initial population is used each time different artificial bee colony algorithms are run independently to emphasize the equity of inter-algorithm comparisons.

The change of global optimal value with number of iterations for the four test problems during the optimization process are shown in Fig. 7, and the ABC2 and HyVABC algorithms have

significant advantages in dealing with high-dimensional problems. The ABC2 algorithm possesses the fastest convergence speed, which is because all bees are guided through the global optimal value for updating. Although the HyVABC algorithm sacrifices a little convergence speed compared to the ABC2 algorithm, it shows remarkable optimization performance on both high and low dimensional problems and successfully obtains the optimal values for the four test problems. Meanwhile, the HyVABC algorithm can still maintain a fairly fast convergence speed because the update of the scouts is guided by the global optimal value. By comparing the optimization process of different algorithms on these four test problems, the strong optimization performance, fast convergence speed and high robustness of the HyVABC algorithm on different test problems are further highlighted.

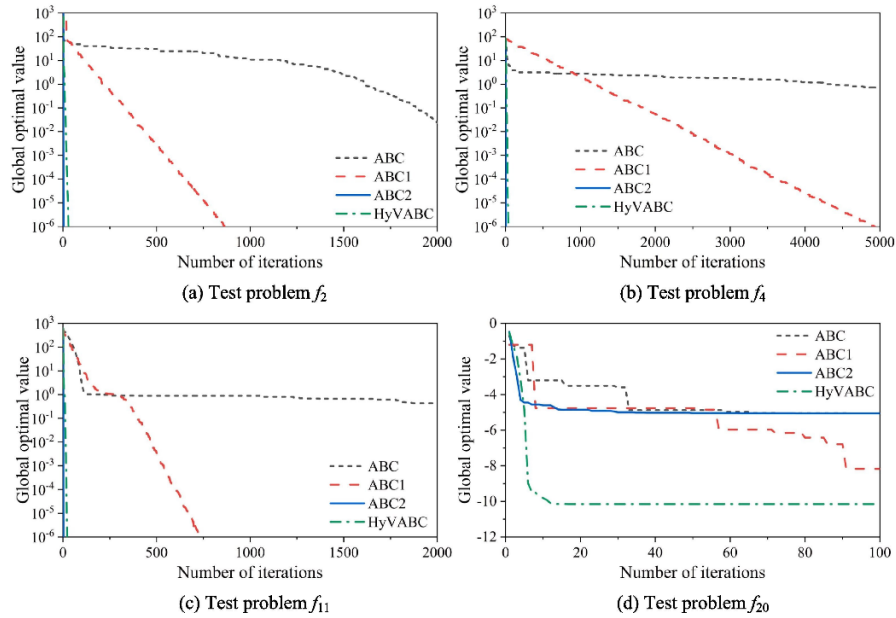


Fig. 7. Change of global optimal value with number of iterations for the four test problems.

3.3 Effects of combinatorial prediction models

To better explore the performance of three combinatorial prediction models in optimization, the exploration and exploitation ratios during the evolution process are analyzed in Fig. 8. Exploration and exploitation are complementary in the HyVABC algorithm. The exploration phase uses random search by individual bees to explore the solution space and avoid local optima, enhancing global search ability. In the exploitation phase, the algorithm focuses on refining high-quality solutions, using information transfer and cooperation to converge toward the global optimum. The exploitation phase's cooperative nature improves local search ability, speeding up convergence. The XGBoost-HyVABC model has the highest exploitation ratio, helping it use known good

solutions more efficiently and accelerating convergence. Throughout the process, exploration and exploitation alternate, forming an iterative optimization that strengthens both global and local search. In the early stages, exploration is more random and extensive, while as iterations progress, exploitation increases to focus on high-quality regions. This dynamic adjustment allows the HyVABC algorithm to balance global and local search effectively at different stages.

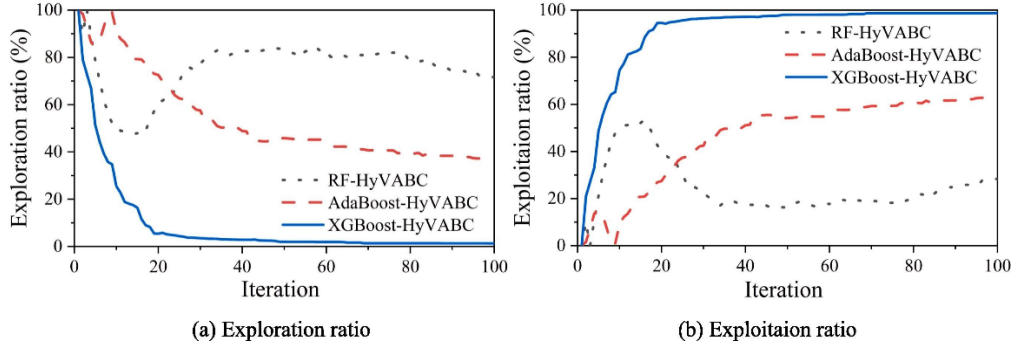


Fig. 8. Exploration ratio and exploitation ratio of three combinatorial prediction models.

Fig. 9 demonstrates the variation of the global optimal value for the MSE on the test set with the number of iterations during the optimization process. The global optimal values of the three models first decrease and then stabilize as the iteration number increases, which shows that the combinatorial prediction model constantly tries to find better hyperparameters to make the ensemble learning model have improved prediction performance and generalization ability. Among the three combinatorial prediction models, the global optimal values of the XGBoost-HyVABC model are always lower than the other two models in all iterations, so the XGBoost-HyVABC model exhibits the strongest generalization ability and prediction performance.

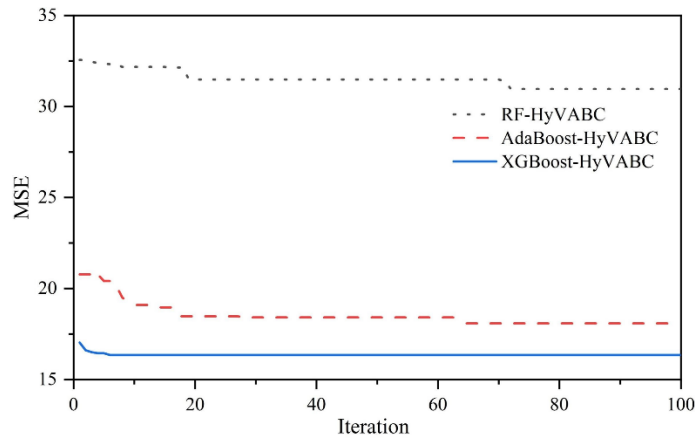


Fig. 9. Variation of the global optimal value for MSE on test set.

The training time per iteration for the three combinatorial prediction models is displayed in Fig.

10, in which the XGBoost-HyVABC model has the least training time per iteration among all three models. In contrast, the overall curve of training time for the AdaBoost-HyVABC model roughly overlaps with that of the RF-HyVABC model, but the fluctuation of training time is larger in certain iteration. As a result, the XGBoost-HyVABC model not only has the strongest prediction performance and generalization ability, but also the shortest training time. Although the training time of the AdaBoost-HyVABC model is roughly comparable to that of the RF-HyVABC model, its prediction performance after convergence is better than that of the RF-HyVABC model. The detailed parameters of the three combinatorial prediction models after convergence are given in Table 3. Considering the prediction performance, generalization ability and training time, the XGBoost-HyVABC model is selected as a proxy model to characterize mapping relationship between key operating parameters and performance parameters of the fuel cell air compressor.

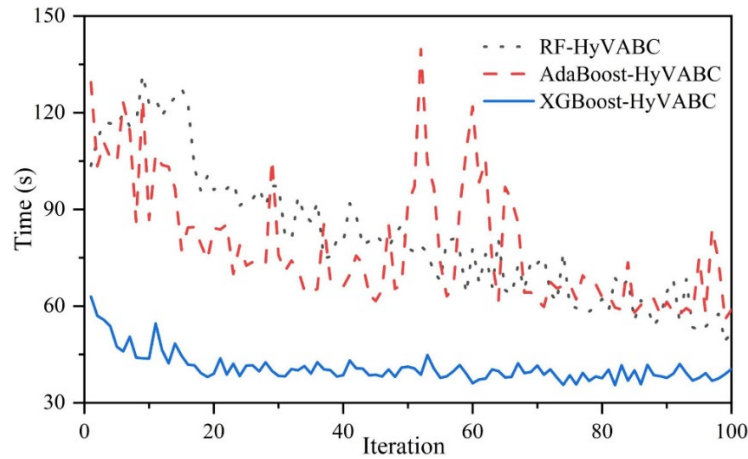


Fig. 10. Training time per iteration for the three combinatorial prediction models.

Table 3. Detailed parameters of the three combinatorial prediction models after convergence.

Model	N_C	M_D	ε	M_L	MSE of test set	Training time(s)
RF-HyVABC	29	24	-	50	31.0	64
AdaBoost-HyVABC	35	13	0.65	-	18.1	62
XGBoost-HyVABC	76	30	0.91	-	16.4	46

The comparison of predicted and test values of the XGBoost-HyVABC model on test set is demonstrated in Fig. 11. The XGBoost-HyVABC model after training exhibits excellent generalization ability and prediction performance, which achieves accurate prediction of flow rate and pressure ratio for the compressor on test set. The absolute values of relative errors between predicted and test values for flow rate and pressure ratio are no more than 3 %, indicating that the

combinatorial prediction model can effectively portray the mapping relationship between operating parameters and performance parameters of the compressor. Meanwhile, the prediction effect of the XGBoost-HyVABC model on pressure ratio is slightly better than that of flow rate, and the distribution of the relative error for the pressure ratio is closer to 0. This is because the order of magnitude and the range of variation for the pressure ratio are smaller than that of the flow rate, and thus the prediction of the pressure ratio is relatively less difficult. Nevertheless, the combinatorial prediction model demonstrates excellent prediction effects for both pressure ratio and flow rate.

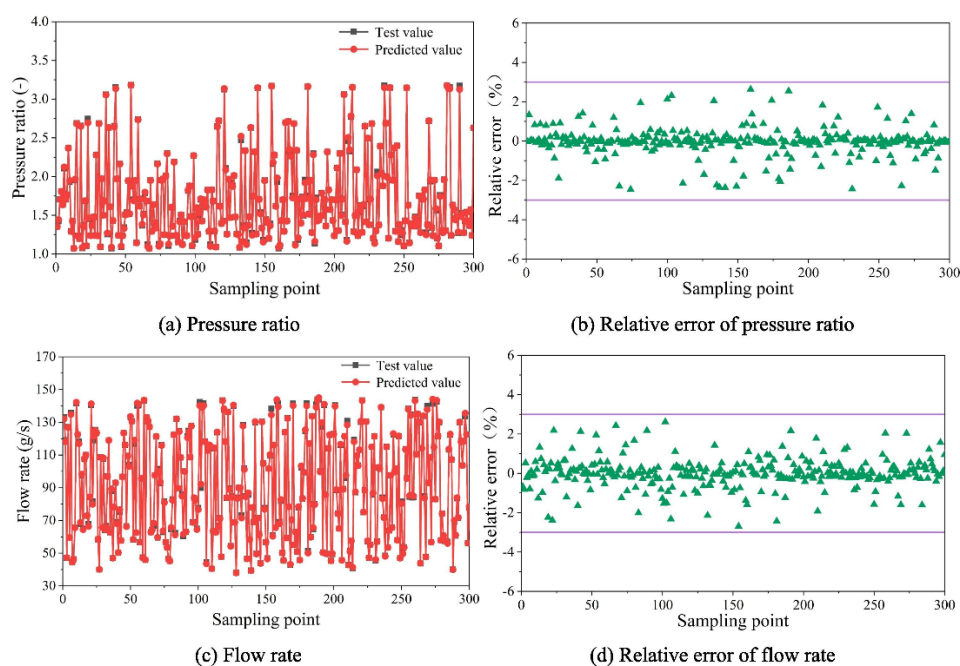


Fig. 11. Comparison of predicted and test values of XGBoost-HyVABC model on test set.

4 Conclusions

In this research, significant efforts have been carried out to realize the performance prediction of the fuel cell air compressor with air bearings considering accelerated decay characteristics. Accelerated endurance tests of the compressor were conducted based on the designed operating conditions, and the HyVABC algorithm was proposed to achieve the enhancement and balance of global optimization capability and local feature capture ability. On this basis, combinatorial prediction models were developed to accurately predict the performance of the compressor. The key findings are summarized below.

- (1). Through 60 h accelerated endurance test, the performance of the compressor exhibits a significant decay, and the decay degree initially increases and then decreases as the speed increases. The decay degree of pressure ratio for the compressor is the greatest at a flow rate of 133 g/s when

the speed is 100000 rpm, with the decay rate of 25.9 % compared with that before break-in.

(2). Considering the prediction performance, generalization ability and training time, the XGBoost-HyVABC model is chosen as a proxy model to characterize mapping relationship between key operating parameters and performance parameters of the centrifugal air compressor. The absolute values of relative errors between predicted and test values for the flow rate and pressure ratio are no more than 3 %.

These findings not only provide a solid algorithmic and modeling foundation for accurately and efficiently predicting the accelerated decay performance of compressors, but also can be applied to structural design and performance optimization of centrifugal air compressors. However, this study only tested for conventional working conditions and did not consider extreme working conditions such as low temperature environment and high altitude. Meanwhile, due to the limitation of the test bench, it was unable to integrate the centrifugal air compressor with the fuel cell system. In the future, multi-objective optimization of compressor performance under complex real-world conditions will be considered for accelerated decay characteristics.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Thanks to Professor Wang Hu for teaching gradient optimization, Gaussian modeling methods, least squares modeling methods, BP and CNN neural network methods in class. The basic theories were explained in great detail, and I learned a lot.

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